

Education

Ph.D. Materials Science and Engineering , <i>University of Wisconsin-Madison</i>	(GPA: 3.70/4.0)	Aug 2024
M.S. Materials Science and Engineering , <i>University of Wisconsin-Madison</i>	(GPA: 3.70/4.0)	Dec 2020
B.S. Engineering , <i>Fort Lewis College</i>	(GPA: 3.99/4.0)	Dec 2017

Skills

Python, PyTorch, scikit-learn, Bash, Git, $\text{\LaTeX}$, OpenHPC, Vim, Docker, Linux, VASP, LAMMPS, Apptainer, C++, MATLAB, AutoCAD, SolidWorks, Verilog, SPICE

Technical Experience

Postdoctoral Appointee **Dec 2024 — Present**
Center for Integrated Nanotechnologies, Sandia National Laboratories Albuquerque, NM

- **Graph Neural Networks:** Built graph neural networks to predict metal-organic framework properties with uncertainty estimates.
- **Bayesian Optimization:** Screened metal-organic frameworks with graph neural network surrogates for favorable separation properties of volatile organic compounds.
- **Microstructure Screening:** Screened LLZO microstructures using Bayesian optimization to identify solid electrolytes with favorable properties.
- **Surface Binding Energies:** Analyzed how surface orientation impacts binding energies and their downstream effects on sputtering.

Research Assistant **Jun 2018 — Aug 2024**
Computational Materials Group, UW-Madison Madison, WI

- **Machine Learning Domain of Applicability for Materials:** Developed a method using kernel density estimation to assess the applicability domain of machine learning models. The method effectively distinguishes chemically distinct groups and relates high dissimilarity with poor model performance and unreliable uncertainty estimates.
- **Scientific Cluster Construction and Administration:** Assisted in building and managing two high-performance computing clusters. Compiled materials research software and implemented Environment Modules to streamline software management.
- **Quantifying Metallic Glass Forming Ability:** Developed a high-throughput workflow for efficiently training machine learning interatomic potentials and simulating complex material properties. Properties were used to develop models for metallic glass forming ability and followed physically rational trends observed in previous research.

Summer Undergraduate Research Experience **May 2017 — Aug 2017**
Computational Nuclear Engineering Research Group, UW-Madison Madison, WI

- **Developed Python Tool:** Automated the visualization of Direct Accelerated Geometry Monte Carlo (DAGMC) geometries in VisIt.

- **Collaboration Friendly Code:** The Python PEP 8 coding style was adopted to facilitate easier reading.

Capstone Design Project

Sep 2016 — Apr 2017

Undergraduate Research, Fort Lewis College Durango, CO

- **Built Instrumentation:** Team designed and built an exotic propulsion test stand.
- **MATLAB Modeling:** Implemented electrostatic displacement mechanism and modeled system response.

Summer Undergraduate Research Fellowship

May 2016 — Aug 2016

Advanced Diagnostics and Propulsion Research Laboratory, Purdue West Lafayette, IN

- **Experimental Imaging:** Operated pressure vessels, X-ray tube sources, and high-speed cameras for analysis of two dimensional sprays.
- **Lead Shielding:** Constructed a protective lead enclosure for X-ray tube sources to shield operating personnel from excessive radiation exposure.

Design Project

Dec 2015 — Apr 2016

Undergraduate Research, Fort Lewis College Durango, CO

- **Sensor Package:** Designed and developed an interchangeable sensor package for measurement of water temperature, oxygen reduction potential, pH, time, and global positioning system data.
- **Simple Deployment:** Package designed to be durable, waterproof, and easy to use by attaching to rafts or kayaks.

Publications

- Lane E. Schultz, Nickolas Gantzler, N. Scott Bobbitt, Dorina F. Sava Gallis, and Rémi Dingreville. “Graph neural network-based multi-objective Bayesian optimization for enhanced screening of metal–organic frameworks with optimal separation performance”. In: *J. Mater. Chem. A* (2026). DOI: [10.1039/D5TA09133K](https://doi.org/10.1039/D5TA09133K)
- Lane E. Schultz, Yiqi Wang, Ryan Jacobs, and Dane Morgan. “A General Approach for Determining Applicability Domain of Machine Learning Models”. In: *npj Computational Materials* (2025). DOI: [10.1038/s41524-025-01573-x](https://doi.org/10.1038/s41524-025-01573-x)
- Lane E. Schultz, Benjamin Afflerbach, Paul M. Voyles, and Dane Morgan. “Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization”. In: *Journal of Materiomics* (2024). DOI: [10.1016/j.jmat.2024.100964](https://doi.org/10.1016/j.jmat.2024.100964)
- Lane E. Schultz, Benjamin Afflerbach, Izabela Szlufarska, and Dane Morgan. “Molecular dynamic characteristic temperatures for predicting metallic glass forming ability”. In: *Computational Materials Science* (2022). DOI: [10.1016/j.commatsci.2021.110877](https://doi.org/10.1016/j.commatsci.2021.110877)
- Lane E. Schultz, Benjamin Afflerbach, Carter Francis, Paul M. Voyles, Izabela Szlufarska, and Dane Morgan. “Exploration of characteristic temperature contributions to metallic glass forming ability”. In: *Computational Materials Science* (2021). DOI: [10.1016/j.commatsci.2021.110494](https://doi.org/10.1016/j.commatsci.2021.110494)
- Jun Meng, Md Sariful Sheikh, Lane E. Schultz, William O. Nachlas, Jian Liu, Maciej P. Polak, Ryan Jacobs, and Dane Morgan. “Ultrafast Oxygen Conduction in Sillén Oxychlorides”. In: *Chemistry of Materials* (2025). DOI: [10.1021/acs.chemmater.4c03163](https://doi.org/10.1021/acs.chemmater.4c03163)

- Vidit Agrawal, Shixin Zhang, Lane E. Schultz, and Dane Morgan. “Accelerating ensemble uncertainty estimates in supervised materials property regression models”. In: *Computational Materials Science* (2025). DOI: [10.1016/j.commatsci.2024.113494](https://doi.org/10.1016/j.commatsci.2024.113494)
- Ryan Jacobs, Lane E Schultz, Aristana Scourtas, KJ Schmidt, Owen Price-Skelly, Will Engler, Ian Foster, Ben Blaiszik, Paul M Voyles, and Dane Morgan. “Machine learning materials properties with accurate predictions, uncertainty estimates, domain guidance, and persistent online accessibility”. In: *Machine Learning: Science and Technology* (2024). DOI: [10.1088/2632-2153/ad95db](https://doi.org/10.1088/2632-2153/ad95db)
- KJ Schmidt, Aristana Scourtas, Logan Ward, Steve Wangen, Marcus Schwarting, Isaac Darling, Ethan Truelove, Aadit Ambadkar, Ribhav Bose, Zoa Katok, Jingrui Wei, Xiangguo Li, Ryan Jacobs, Lane Schultz, Doyeon Kim, Michael Ferris, Paul M. Voyles, Dane Morgan, Ian Foster, and Ben Blaiszik. “Foundry-ML - Software and Services to Simplify Access to Machine Learning Datasets in Materials Science”. In: *Journal of Open Source Software* (2024). DOI: [10.21105/joss.05467](https://doi.org/10.21105/joss.05467)
- Ryan Jacobs, Maciej P. Polak, Lane E. Schultz, Hamed Mahdavi, Vasant Honavar, and Dane Morgan. “Regression with Large Language Models for Materials and Molecular Property Prediction”. In: *Digital Discovery* (2024). DOI: [10.48550/arXiv.2409.06080](https://doi.org/10.48550/arXiv.2409.06080)
- Benjamin T. Afflerbach, Carter Francis, Lane E. Schultz, Janine Spethson, Vanessa Meschke, Elliot Strand, Logan Ward, John H. Perepezko, Dan Thoma, Paul M. Voyles, Izabela Szlufarska, and Dane Morgan. “Machine Learning Prediction of the Critical Cooling Rate for Metallic Glasses from Expanded Datasets and Elemental Features”. In: *Chemistry of Materials* (2022). DOI: [10.1021/acs.chemmater.1c03542](https://doi.org/10.1021/acs.chemmater.1c03542)
- Benjamin T. Afflerbach, Lane Schultz, John H. Perepezko, Paul M. Voyles, Izabela Szlufarska, and Dane Morgan. “Molecular simulation-derived features for machine learning predictions of metal glass forming ability”. In: *Computational Materials Science* (2021). DOI: [10.1016/j.commatsci.2021.110728](https://doi.org/10.1016/j.commatsci.2021.110728)
- J. Xi, G. Bokas, L.E. Schultz, M. Gao, L. Zhao, Y. Shen, J.H. Perepezko, D. Morgan, and I. Szlufarska. “Microalloying effect in ternary Al-Sm-X (X=Ag, Au, Cu) metallic glasses studied by ab initio molecular dynamics”. In: *Computational Materials Science* (2020). DOI: [10.1016/j.commatsci.2020.109958](https://doi.org/10.1016/j.commatsci.2020.109958)
- B.R. Halls, J.R. Gord, L.E. Schultz, W.C. Slowman, M.D.A. Lightfoot, S. Roy, and T.R. Meyer. “Quantitative 10-50 kHz X-ray radiography of liquid spray distributions using a rotating-anode tube source”. In: *International Journal of Multiphase Flow* (2018). DOI: [10.1016/j.ijmultiphaseflow.2018.07.014](https://doi.org/10.1016/j.ijmultiphaseflow.2018.07.014)

Presentations

- Lane E. Schultz, Nickolas Gantzler, N. Scott Bobbitt, Dorina F. Sava Gallis, and Rémi Dingreville. “GNNs for Selection of Optimal Metal Organic-Frameworks”. In: Electronic Materials Conference. Ann Arbor, MI, 2026
- Lane E. Schultz, Nickolas Gantzler, N. Scott Bobbitt, Dorina F. Sava Gallis, and Rémi Dingreville. “GNNs for Selection of Optimal Metal Organic-Frameworks”. In: The Minerals, Metals & Materials Society. San Diego, CA, 2026

- Lane E. Schultz, Nickolas Gantzler, N. Scott Bobbitt, Dorina F. Sava Gallis, and Rémi Dingreville. “GRAPH NEURAL NETWORK-DRIVEN MULTI-OBJECTIVE BAYESIAN OPTIMIZATION FOR DISCOVERING METAL-ORGANIC FRAMEWORKS WITH OPTIMAL SEPARATION PERFORMANCE.”. In: Chesapeake Large Scale Analytics Conference. Annapolis, MD, 2025
- Lane E. Schultz, Benjamin T. Afflerbach, and Dane Morgan. “Molecular Dynamic Characteristic Temperatures for Predicting Metallic Glass Forming Ability”. In: Materials Science & Technology. Columbus, OH, 2021
- Lane E. Schultz, Benjamin T. Afflerbach, and Dane Morgan. “Molecular Dynamics Features for Predicting Metallic Glass Critical Casting Thickness”. In: Virtual Materials Research Society Spring/Fall Meeting & Exhibit. 2020
- Lane E. Schultz, Thomas J. Cogger, Ryan Good, James Schneider, Robert Rothschild, and William Nollet. “Design of Torsional Test Stand for Micro-Newton Force Detection”. In: 2018 Aerodynamic Measurement Technology and Ground Testing Conference. 2018. DOI: [10.2514/6.2018-3737](https://doi.org/10.2514/6.2018-3737)
- James Schneider, Lane E. Schultz, Sophie Mancha, Eric Hicks, and Ryan N. Smith. “Development of a portable water quality sensor for river monitoring from small rafts”. In: OCEANS 2016 MTS/IEEE Monterey. 2016. DOI: [10.1109/OCEANS.2016.7761392](https://doi.org/10.1109/OCEANS.2016.7761392)

Teaching Experience

Assisting undergraduates with research	Undergraduate
Assisting laboratory group peers with software installation and cluster use	Graduate
Assisted in class labs (molecular dynamics)	Graduate
Grader for Thermodynamics of Solids	Graduate
Public school substitute teacher	High/Middle/Elementary
Teaching assistant for Thermal and Fluid Systems Laboratory	Undergraduate
Teaching assistant for Engineering Fundamentals II (MATLAB)	Undergraduate

Awards, Honors, and Societies

<i>PPG Fellowship</i>	University of Wisconsin-Madison, Madison, WI
<i>Ying Yu Chuang Graduate Support Award</i>	University of Wisconsin-Madison, Madison, WI
<i>Sigma Pi Sigma (Physics Honor Society)</i>	Fort Lewis College, Durango, CO
<i>Order of the Engineer</i>	Fort Lewis College, Durango, CO
<i>Deans' Council Freshman 4.0 Award and Certificate</i>	Fort Lewis College, Durango, CO
<i>Freshman Chemistry Recognition Award</i>	Fort Lewis College, Durango, CO
<i>Renaissance Plaque</i>	Manzano High School, Albuquerque, NM
<i>Chickasaw Honor Club Outstanding Academic Achievement Award</i>	Menard High School, Menard, TX
<i>Patrick S. Gilmore Band Award</i>	Menard High School, Menard, TX

Additional Information

Citizenship	United States, Colombia, and the Chickasaw Nation
Languages	English and Spanish (fluent written and verbal)